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Reanalyzing frustration: Event maximality and inertia in two O'dam frustratives*

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Abstract *Frustrative* particles modify a clause by conveying the non-realization of an associated expectation. The semantics of frustrativity are relatively understudied, complicated by variation in whether the relationship between the marked clause and the 'target' of frustration is lexically or only pragmatically specified. We contribute to the emerging picture by reporting on O'dam, which is typologically unusual in having two contrastive frustratives: *tɨɨ* and *tɨɨpup*. Where most existing analyses treat frustratives as clausal markers of *non-inertia*, we argue that O'dam frustratives are also sensitive to event-based *non-maximality*. We propose a unified approach on which the differences are derived from a parametric presuppositional contrast: the weaker *tɨɨ* commits its speaker to the non-inertial status of the evaluation world, while *tɨɨpup* imposes a stronger form of non-inertia by blocking maximal instantiation of the marked-clause event. Our analysis suggests that the non-inertial approach is appropriate for weak frustratives (which do not fully preclude 'frustrated' outcomes; Kroeger 2024), but opens the door to a reanalysis of strong frustrativity (Copley & Harley 2014; Davis & Matthewson 2022: a.o.) in the crosslinguistic landscape.

Keywords: Frustratives, inertia, modality, aspect, telicity, event maximality, O'dam

1 Introduction

Frustrative particles belong to a class of linguistic devices that can be used to signal that a situation has developed against expectation or intent. Defined as elements "that express the non-realization of some expected outcome [associated with] the marked clause," (Overall 2017: 479), frustratives are the subject of a growing body of work (see, a.o., Copley & Harley 2014; Davis & Matthewson 2016; Carol & Salanova 2017; Kroeger 2017): their semantic analysis is of particular interest because their

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range of uses overlaps with clausal counter-to-expectation markers (e.g., *concessive* conjunctions) as well as subclausal elements (e.g., non-culminating aspects).

Across languages, frustratives may be associated with unrealized intentions or desires as well as with failed expectations, giving them a broadly modal feel. While they formally compose with a single clause (see Overall 2017), they often imply the relevance of a second proposition which acts as the unrealized 'target' of frustration. For instance, (1) (Tohono O'odham; Uto-Aztecan, ISO 639-3 ood) conveys that Juan opened the door, but some anticipated outcome did not occur: perhaps an expected arrival was not there, Juan himself did not depart, or the door did not remain open.

(1)

Huan	'at	**cem**	ku:pio	g	pualt
Juan	aux.PERF	FRST	open	DET	door

'Juan opened the door (in vain).' (Copley & Harley 2014, 29c)

Alongside uses like (1)—in which the target of frustration is external to the marked clause—frustratives can license interpretations which target part or all of the marked-clause eventuality. Recent work (Copley & Harley 2014; Davis & Matthewson 2016, 2022) has argued that this variation is linked to grammatical aspect: frustratives thus operate at the aspect-modality interface, and their precise range of uses varies with language-specific resources for these categories.

This paper contributes to the emerging semantic picture by reporting on O'dam (Uto-Aztecan, ISO 639-3 stp). O'dam is unique in having two particles—*tɨi* (FRST) and *tɨipup* (FRST.NONMAX)—whose contrastive interpretation offers a view into the decompositional components of frustrativity. While existing accounts treat frustratives as propositional markers of *non-inertia* (Dowty 1979), we argue that O'dam frustrativity is also sensitive to subclausal structure: in particular, to a notion of event *maximality* (Filip 2008). We propose a unified approach to the two particles, deriving their differences from a parametric contrast in the encoding of unrealized expectation: the stronger particle *tɨipup* commits its speaker to non-maximal realization of the marked-clause event, while the weaker *tɨi* only commits to the non-inertiality of the evaluation world. This analysis offers potential insight into the crosslinguistic semantic typology of frustrativity, suggesting that frustrative markers vary along at least two dimensions: in the strength of a non-inertial commitment and in the level (clausal or subclausal) at which 'deviation' is invoked.

2 Frustratives across languages: a non-exhaustive list of uses

Frustratives have a diverse range of uses. We list some crosslinguistically common functions below, with examples from Tohono O'odham (TO; Copley & Harley 2014) and Kimaragang (KD; Kroeger 2017, 2024; Dusunic, ISO 639-3 kqr).[1]

In canonical or **proper** uses (cf. Carol & Salanova 2017), frustratives entail the realization of the entire event described by the marked clause, treating an expected or intended outcome of this event (or a salient associated situation; Davis & Matthewson 2022) as the target of frustration. Examples (1)-(2a) are of this type: (2a) makes the target of frustration explicit in an optional second clause. Proper frustrativity is also available when the underlying predicate is stative: (2b) can be used if the speaker is ready, but the event for which they became ready failed to materialize.[2]

(2) a. n-o-sii-∅ ku no **dara** it=tasu (nga' n-iit-an
PST-NVOL-shoo-OV 1SG already FRST NOM=dog (but PST-bite-DV
oku-i')
1SG=EMPH)
'I shooed the dog (but still got bitten).' (KD; Kroeger 2017, 1g)

b. **cem** 'añ ñ-na:tokc
FRST 1SG 1SG-ready
'I was ready (but to no avail).' (TO; Copley & Harley 2014: 17)

Examples (3a)-(3b) illustrate **incompletive** frustrativity. In combination with telic predicates, these examples have an *antiresultative* flavor, conveying that the marked-clause event ended before reaching its natural (lexically-specified) culmination. Incompletive uses thus target proper parts of the marked-clause event.

(3) a. k<um>orop no **dara** it=pilat dialo, naka-raa
<AV>scab COMPL FRST NOM=wound 3SG PST.AV.NVOL-blood
kembagu
again
'His wound was beginning to scab over, but then it started bleeding again.' (KD; Kroeger 2024: 5b)

b. Huan 'o **cem** kukpi'ok g pualt.
Juan aux.IMPF FRST open DET door
'Juan was trying to open the door.' (TO; Copley & Harley 2014: 29b)

1 See Overall (2017) for discussion of a wider range of frustrative functions and their relationship to other types of counter-to-expectation expressions.

2 Copley & Harley (2014) also note a second, **cessative**, interpretation for (2b), on which it conveys that the speaker was ready but is now no longer ready. Cessative readings are sometimes treated as distinct from proper frustrativity; insofar as (1) also has an interpretation in which frustration targets the continuation of the lexically-specified result state of the underlying telic predicate, we follow Copley & Harley in classifying them together. The underlying intuition is that the continuation of a state may in certain contexts be treated as a relevant consequence thereof.

Frustratives can also target the full marked clause event, as in (4). **Avertive** readings are reported to be felicitous just in case something is going on in the reference context which might be expected to lead to the realization of the described event: in this sense, they are reasonably well-paraphrased with prospective (proximate-future) markers like *be going to* or *be about to*. In some languages, avertive readings require overt futurate marking, but (4a) shows that this is not universal (see Kroeger 2024).[3]

(4) a. iit-an oku no **dara** da-tasu nga' a=tanak po=ot
bite-DV 1SG COMPL FRST GEN=dog but NOM=child FOC=NOM
nokoponii
AV.PST.say.*sii*

'I was about to be bitten by the dog, but the child said "Shii!" (the dog did not bite me) (KD; Kroeger 2024: 6b)

b. Huan 'at o **cem** kukpi'ok g pualt
Juan aux.PERF PROSP FRST open DET door.

'Juan was going to open the door.' (He tripped on the way to the door) (TO; Copley & Harley 2014: 29a)

In some languages, frustratives can also be **optative**, marking a modified but as-yet unrealized proposition as desirable to the speaker or subject: (5a) is differentiated from a standard avertive in leaving open the possibility that the desired event will still come to pass. Kroeger reports that Kimaragang *dara* can also mark a declarative as a polite request: (5b) seems most closely related to incompletive readings.

(5) a. mongolos oku **dara** dit=widio dialo
AV.borrow 1SG FRST ACC=video 3SG

'I would like to borrow his video machine.' (KD; Kroeger 2024: 10c)

b. ∅-po-owit oku **dara** dikaw do=sigup
AV-CAUS-bring 1SG.NOM FRST 2SG ACC=tobacco

'Please bring me a little tobacco.' *lit:* I cause you to bring tobacco *dara* (KD; Kroeger 2024: 11a)

Finally, frustratives are frequently found in counterfactual conditionals, but their distribution in these contexts and any interaction with independent markers of counterfactuality has yet to be fully explored. While we do find optative, counterfactual, and politeness uses for O'dam frustatives, we focus here on the three 'core' uses in (2)-(4), only offering some comments on other uses in Section 5.3.

3 To the extent that some form of futurate marking—whether overt, covert, or coerced (cf. Kroeger 2024) is required for avertive frustrativity, it must be prospective. Davis & Matthewson (2022) show that frustrative interpretations actually provide a diagnostic for differentiating between prospectivity and true (modal) futurity in St'át'imcets.

3 Frustrative particles in O'dam: *tɨi* and *tɨipup*

Spoken in the Mexican states of Durango, Nayarit, and Zacatecas, O'dam is a Southern Tepehuan language of the Tepiman subbranch of the Uto-Aztecan language family (García Salido & Everdell 2020).[4] The language is highly agglutinating; as in similar languages, the only obligatory clause element is the verb. The internal structure of most larger phrases (e.g., DPs, PPs) is rigid, but the ordering of these phrases within a clause is relatively free. We refer the reader to García Salido (2014, 2021) and Everdell (2023) for further details: here we simply note that both frustratives appear in the preverbal position, otherwise occupied by clause-modifying particles. O'dam frustratives thus appear in a position suggestive of clausal scope.

There is limited prior description of *tɨi* and *tɨipup*: Willett (1991) glosses both particles as markers of *non-realized intention*, which is broadly compatible with the frustrative definition in Section 1. We find, however, that the two particles differ in use: while all of the functions in Section 2 are attested, they are not equally shared.[5]

Both *tɨi* and *tɨipup* can convey incompletive meanings, but not interchangeably. Using *tɨi*, (6a) can be uttered by a speaker who is still waiting for the bus: she conveys that it is late, but does not rule out that it may still arrive. By contrast, (6b) can only be used if the speaker has given up—i.e., if the waiting ended unsuccessfully.

(6) a. añ **tɨi** nɨɨra-' gu camion
1SG.SBJ FRST wait-IRR DET bus.
'I am waiting for the bus.' (but it still has not come)

b. añ **tɨipup** nɨɨra-t gu camion
1SG.SBJ FRST.NONMAX wait-IMPF DET bus
'I was waiting for the bus.' (but it never arrived)

A similar distinction emerges for the avertive claims in (7): (7a) can be used if buying chips is still an option, but (7b) cannot be used by a speaker who might still go to the store.

4 All O'dam data presented in this paper were collected via fieldwork conducted by the second author. A selected list of glosses: CMP 'completive', CONTR 'counter-expectation', DES 'desiderative', DIR 'directional', MOV 'movement', PNCT 'punctual', PO 'primary object', VIZ 'visual'. For all others refer to the Leipzig Glossing Rules.

5 Despite their suggestive similarity, we remain agnostic regarding a synchronic morphological relationship between *tɨi* and *tɨipup*. Willett & Willett (2015: 147) propose that an independent *pup* particle encodes absence of intention, but provide few empirical examples thereof: we do not find the existing data sufficiently informative to commit to a particular semantic account of *pup*, and our consultants have so far rejected its use in novel contexts. We thus have no clear evidence which adjudicates against treating both O'dam frustratives as monomorphemic. Nevertheless, we aim to develop an analysis on which *tɨipup* is strictly stronger than *tɨi*, so that our approach remains compatible with the idea that the former corresponds to the composition of the latter with a (possibly bound) *pup* morpheme (cf. Koontz-Garboden's 2007 *Monotonicity Hypothesis*).

(7) a. (cham bia'-iñ gu popotes,) **tɨɨ**
(NEG have-1SG.SBJ DET chips) FRST
ba-ja-saba'n-mɨra-k-añi-ch mu tienda
CMP-3PL.PO-buy-MOV-PNCT-1SG.SBJ-PFV DIR store
'(I don't have chips,) I was going to buy them at the store' (but I turned around)

b. **tɨɨpup** jii-ñi-ch mu tienda
FRST.NONMAX go.PFV-1SG.SBJ-PFV DIR store
'I almost went to the store' (but I never even left and now I won't/can't go)

The particles thus differ in whether or not the target of frustration can still be realized at speech time: *tɨɨ* leaves this possibility open, while *tɨɨpup* closes it off. This contrast is linked to a pattern in the default temporal orientation of O'dam frustrative claims. Even absent overt temporal marking, we find that *tɨɨpup* shows a strong preference for past interpretation which is not shared by *tɨɨ*: this is explained if *tɨɨpup* requires its speaker to know how things turned out, since this information is only available when the relevant events (and their consequences) are in the past.

An additional point of contrast emerges when we consider proper frustrativity. The *tɨɨ* example in (8a) is similar to (1) and (2): it conveys that the addressee has killed the snake, but that subsequent events did not go as expected. *Tɨɨpup* is also permitted here, but not on the proper reading: (8b) is infelicitous if the snake has been killed, but is acceptable if the snake is merely injured or if it escaped unharmed.

(8) a. ap **tɨɨ** mua dhi'-ñi ko'
2SG.SBJ FRST kill.SG DEM.PROX-VIZ snake
'You killed this snake (in vain).' (someone else took it to eat it)

b. ap **tɨɨpup** mua dhi'-ñi ko'
2SG.SBJ FRST.NMAX kill.SG DEM.PROX-VIZ snake
Proper (intended, unavailable): '#You killed this snake (in vain).'
Incompletive: 'You tried to kill this snake (but only hurt it).'
Avertive: 'You were about to kill this snake (but it escaped).'

As in the examples in (3), incompletive uses of *tɨɨpup* with telic predicates seem to single out the *culmination condition* of the input predicate as the frustrative target, so that—looking only at (6b) or the incompletive us of (8b)—we might be tempted to characterize *tɨɨpup* as an antiresultative progressive. However, closer consideration of the interaction between *tɨɨpup* and the aspectual class of the marked-clause predicate shows that the effect is better described as one of *non-maximality*. A purely antiresultative meaning would predict the impossibility of incompletive readings where *tɨɨpup* modifies a clause containing an atelic predicate, but such readings are indeed available, as long as the predicate can be partially realized. In (9a), for instance, the speaker partially realizes a seeing-Marcelo event: they convey that some part of Marcelo remained obscured, and the use of the frustrative additionally signals that this was not originally intended or expected. In contrast, incompletive frustrativity with *tɨipup* is impossible when the underlying predicate cannot be partially realized, regardless of aspectual class: both the achievement and activity predicates in (9b)-(9c) license only avertive interpretations.[6]

(9) a. xib **tɨipup** tɨi-ñi-ch gu marcelo
today FRST.NONMAX see.PFV-1SG.SBJ-PFV DET Marcelo

Incompletive: 'I partially saw Marcelo today.' (e.g., through a fence)

b. **tɨipup** na=ñ chu-mataimda-'
FRST.NONMAX SUB=1SG.SBJ DUR-nixtamalize-IRR

Only avertive: 'I was gonna nixtamalize (corn)'[7]

c. Context: some kids tied up the dog and then got it excited

ya' **tɨipup** mɨi gu gagoox
DIR FRST.NONMAX run.SG.PFV DET dog

Only avertive: 'The dog tried to run (but was not able to take even a step).'

Tɨipup is thus more restricted than *tɨi* in two respects. First, it imposes a non-maximality requirement on the marked-clause predicate, restricting the target of frustration to what is described in this clause and ruling out 'proper' readings. Second, *tɨipup* rules out the possibility of a "better" outcome, in which the frustrated target is ultimately realized. Informally speaking, then, we find that O'dam speakers can choose between two degrees of frustrativity: a strong form associated with *tɨipup*, and a weaker (less committal) form using *tɨi*.

4 Inertia, efficacy, and commitment

We now briefly review some previous approaches to frustrative semantics. While we do not find an existing account which can be immediately adapted to the empirical data discussed above, we draw on several insights from the existing literature in developing descriptive generalizations which form the basis of our formal proposals.

6 For some additional discussion and data, we refer readers to Everdell & Nadathur (2025).

7 Nixtamalization refers to the preparation of corn (e.g., for use in tortillas) by mixing it with lime. Our consultants treat this as an achievement: the corn is nixtamalized from the moment that it is mixed with lime, so no partial realization is possible. In order to express that something has gone wrong with nixtamalization, speakers choose *tɨi* over *tɨipup*: *tɨi* allows a reading on which nixtamalization occurs, but its results are not as expected or desired.

4.1 Frustrativity and inertial modality

As Carol & Salanova (2017) point out, if frustratives are taken to interact directly with an event predicate in the marked clause, incompletive and avertive readings are much more easily analyzed than proper ones. Since both interpretations target a lexically-specified part of the input predicate, they can be analyzed using tools developed for handling non-culminating aspects: namely, a Dowty-inspired form of *inertial modality* which associates an actual situation with its maximally normal or stereotypical continuations. Per (10), adapted from Carol & Salanova (2017: 6), frustratives require that the inertial alternatives to a reference situation s instantiate the input predicate P while precluding an actual P-event, making s itself non-inertial.

(10) $[\![\text{FRST}]\!] := \lambda P_{v,st} \lambda s. \forall s' \in \text{INR}(s)[\exists e.P(e)(s')] \,\&\, \neg\exists e.P(e)(s)$

Carol & Salanova suggest that incompletive and avertive frustrativity differ only in the *type* of inertia for which they select (see Arregui, Rivera & Salanova 2014 for discussion; also Tatevosov 2008 for a similar treatment). Incompletive frustrativity employs *event-based* inertia, associating situations in which a P-event has begun with situations in which this event continues uninterrupted: where P is telic, the second conjunct of (10) blocks actual culmination. Avertivity instead relies on *preparatory* inertia, associating situations which realize preparations for a P-event with those in which these preparations continue uninterrupted to P. However, the second conjunct of (10) makes it incompatible with proper readings, meaning that this approach necessitates a polysemous treatment of frustratives—like Tohono O'odham *cem*, Kimaragang *dara* and O'dam *tɨi*—which allow all three readings: proper frustrativity must establish the actual occurrence of a P-event while simultaneously introducing a second event which follows inertiall from P but goes unrealized in the actual world.

Working on languages (Tohono O'odham and St'át'imcets, respectively) in which aspect is clearly indicated, Copley & Harley (2014) and Davis & Matthewson (2016, 2022) argue that the polysemy problem is in fact illusory, and can be resolved by considering the aspectual specification of a frustrative clause: the distribution of proper, incompletive, and avertive frustrative interpretations in both languages corresponds to the distribution of *perfective, imperfective,* and *prospective* aspect. This allows frustratives to be treated purely propositionally: in each case, the frustrative establishes the occurrence of a completed, ongoing, or upcoming situation (as described by the inflected clause), imposing the additional constraint that some normal consequence of this situation does not obtain in the actual world. Whether this consequence is part (or all) of the underlying (uninflected) predicate or altogether external is determined not by pragmatic context, but by the aspectual inflection of the input proposition p.

(11) $[\![\text{FRST}]\!] := \lambda w \lambda p_{st} . p(w) \,\&\, \partial(\exists q_{st}(\neg q(w) \,\&\, \forall w' \in \text{INR}(w)[p(w') \rightarrow q(w')]))$[8,9]

While (11) improves on (10), it is not fully suited to our needs. The mapping between aspectual marking and frustrative flavor which seems appropriate for Tohono O'odham and St'át'imcets is not well-supported in the O'dam data. We find acceptable uses of *tɨipup* in combination with perfective marking (see 7b), despite incompatibility with proper frustrativity: in fact, our consultants prefer perfective interpretations of *tɨipup*-modified clauses. Relatedly, it is not clear that an imperfective-driven notion of incompletion (i.e., non-culmination) can be mapped straightforwardly onto the non-maximality requirement which is relevant for *tɨipup*. Simultaneously, however, in precluding the target (q) of frustration, the first presuppositional conjunct in (11) renders it too strong as a candidate denotation for *tɨi*. More abstractly, while the introduction of a second, unexpressed proposition q is evidently necessary to capture (a strong form of) proper frustrativity on this approach, it is less appropriate for incompletive and avertive uses: since nothing forces q to describe part or all of the event predicate underlying clause p, (11) potentially overgenerates for data like (3), (4), (6), and (7).

4.2 Frustrativity as non-efficacy

Since Tohono O'odham and O'dam are closely related, Copley & Harley's (2014) aspect-sensitive treatment of *cem* may offer a helpful starting point. The shape of Copley & Harley's analysis is shared with (11), but their implementation uses the **force dynamics** framework (Copley & Harley 2015), which obviates the need to reference a second (potentially superfluous) proposition as the target of frustration.

In force dynamics, neo-Davidsonian events are replaced by *forces* (type $\langle s,s \rangle$), conceptualized as deterministic maps from one static situation—an "annotated snapshot of [...] individuals and their [...] properties" (Copley & Harley 2014: 126)—to another. On this approach, a dynamic event predicate becomes a predicate of forces which change one more of the static annotations of an input situation. As shown in (13), a telic VP like $\sqrt{}$*Juan open the door* describes a force which outputs a situation verifying the lexically-specified result state of the force: statives (and propositions) are treated as properties of situations. Copley & Harley assume that the net force of a situation is always recoverable, predicting a unique successor, and

8 (11) is adapted from Davis & Matthewson (2022: 1358). They argue against treating q as a causal consequence of p, suggesting that—at least in St'át'imcets, it is enough that p and q are ordinarily expected to co-occur. We set this issue aside for the time being.

9 The presuppositional status of the frustrative effect in (11) (marked above by ∂; Beaver & Krahmer 2001) is defended by both Copley & Harley and Davis & Matthewson (see also Copley 2005). This assumption also seems appropriate for O'dam based on the obligatory scope of frustratives and negation (*cham*) in the PreV position (see Everdell & Nadathur 2025).

establishing the correspondences in (12).

(12) Let s be a situation and f its net force: $\text{net}(s) = f$

a. *Initial situation of a force:* $\text{init}(f) = \text{net}^{-1}(f) \quad (= s)$

b. *Final situation of a force:* $\text{fin}(f) = f(\text{net}^{-1}(f)) \quad (= f(s)$

c. *Successor of a situation:* $\text{succ}(s) = \text{fin}(\text{net}(s)) \quad (= \text{fin}(f))$

d. *Predecessor of a situation:* $\text{pred}(s) = \text{succ}^{-1}(s)$

(13) a. $[\![\text{Juan open the door}]\!] = \lambda f_{ss}.\text{source}(J, f)\,\&\, [\![\text{open the door}]\!]\,(\text{fin}(f))$

b. $[\![\text{open the door}]\!] = \lambda s.\text{the door is open in } s$

Grammatical aspects operate on force predicates, mapping VPs like (13a) to predicates of situations by specifying a particular situation whose net force is described by the VP. Following Copley & Harley, the *completive* effect of perfectives follows from predicating a VP as the net force of the predecessor of the topic situation s (so that any changes resulting from the VP will obtain in s; see 12d). *Ongoing* imperfectives predicate VPs as the net force of the topic situation itself, and prospectives predicate over the net force of the successor to the topic situation.[10]

(14) a. $[\![\text{PFV}]\!] := \lambda \pi_{ss,ss} \lambda s.\pi(\text{net}(\text{pred}(s)))$

b. $[\![\text{IMPF}]\!] := \lambda \pi \lambda s.\pi(\text{net}(s))$

c. $[\![\text{PROSP}]\!] := \lambda \pi \lambda s.\pi(\text{net}(\text{succ}(s)))$

The special effect of frustrativity, as in (11), is to establish that the topic situation does not develop as expected. Copley & Harley capture this by using a notion of *efficacy*: in the force dynamics, a situation is efficacious just in case its expected successor (i.e., the final result of its net force) actually obtains. Frustrative clauses run counter to expectation by presupposing that their topic situation is not efficacious:

(15) $[\![\text{FRST}]\!] := \lambda s \lambda p_{st}.p(s)$, defined iff s is not efficacious

Combining (12)-(14) with (15) yields the desired interpretations for the proper, incompletive, and avertive frustrative claims in (1), (3b), and (4b), respectively:

(16) a. $[\![(1)]\!]^s = [\![\mathit{cem}(\text{PFV}(\text{Juan open the door}))]\!]^s$
$= \text{source}(J, \text{net}(\text{pred}(s)))\,\&$ the door is open in s

defined iff: some expected result of the open-door situation does not occur (succ(s) does not obtain)

b. $[\![(3b)]\!]^s = [\![\mathit{cem}(\text{IMPF}(\text{Juan open the door}))]\!]$
$= \text{source}(J, \text{net}(s))\,\&$ the door is open in succ(s)

defined iff: the door does not fully open (succ(s) does not obtain)

10 Copley & Harley (2014) argue that PROSP predicates the input force of *some* (not necessarily immediate) successor to the topic situation; we modify (14c) for simplicity.

c. $[\![(4b)]\!]^s = [\![cem(\text{PROSP}(\text{Juan open the door}))]\!]^s$
$= \text{source}(J, \text{net}(\text{succ}(s)))\ \&$ the door is open in $\text{succ}^2(s)$
defined iff: the opening force is never realized ($\text{succ}(s)$ does not obtain)

While this approach is appealing, it too cannot be adopted wholesale for either O'dam frustrative.[11] While it does not require reference to an unstated proposition, (15) suffers from many of the same problems (with respect to the O'dam data) as the Davis & Matthewson-based proposal in (11). Overt aspectual marking appears to align with frustrative flavor as predicted in Tohono O'odham, but the same is simply not true in O'dam: our data suggest that aspect is not obligatorily marked in O'dam, and in any case does not align with the way in which proper vs. incompletive (and avertive) interpretations are differentiated in this language.[12] This suggests that aspectual marking and (in)completion—or rather, (non)maximality—must be decoupled in the eventual O'dam analysis, so that the semantic specification which blocks *tɨipup* in proper uses does not require incompatibility with perfective aspect.

The definition of *non-efficacy* is somewhat imprecise. Based on the deterministic character of forces, a natural interpretation is that the non-efficacy of situation *s* precludes the realization of any of the consequences associated with the net force of *s*; since Copley & Harley treat *cem* as committing to the non-realization of the frustrative target, this interpretation seems intended. This is consistent with the effects of *tɨipup* but too strong for *tɨi*; an alternative interpretation on which non-efficacy simply encodes non-commitment to efficacy is appropriate for *tɨi*, but too weak for *tɨipup*. In either case, non-efficacy is insufficiently granular for O'dam.

Nevertheless, the notion of non-efficacy offers some insight into the right way to capture the intuitive contrast. Crucially, the problem in an inertial formulation like (11) is in its specificity: Davis & Matthewson convey the non-inertial status of the reference situation *s* via a specific commitment to the non-realization of the frustrative target. Efficacy, on the other hand, is a property of situations. Shifting to this type of conception in the modal frame affords us the needed granularity by allowing us to differentiate between a strong speaker commitment against a specific inertial effect and a weak (non-specific) commitment to the non-inertia (*viz.*, non-efficacy) of the actual world: we implement this distinction in Section 5.

4.3 Strong and weak frustrative commitments

We arrive at the following conclusions. Given the distribution of the two O'dam particles and the strength difference between them (as well as their potential morpho-

11 We focus on O'dam-specific issues here, but Davis & Matthewson (2016, 2022) and Kroeger (2017, 2024) discuss some broader crosslinguistic concerns with the force dynamics approach.

12 Per H. Harley, p.c., the aspectual picture presented in Copley & Harley (2014) may be oversimplified.

logical relationship), we would like their differences to emerge from a parametrized semantic contrast: a component of meaning present in the stronger *tɨipup* but absent from the weaker *tɨi*. This contrast necessarily relates to a notion of maximality associated with the underlying event predicate: we thus conclude that—at least in O'dam—frustratives must split the difference between event-based formulations like (10) and propositional approaches like (11) and (15) and take both an aspectual marker and an event predicate as arguments, effectively allowing the frustrative to see 'through' optional aspectual marking to the input predicate. *Tɨi* will then be compatible with both maximal and non-maximal instantiations of the underlying event predicate, permitting the same range of variation that we see for Tohono O'odham *cem* and Kimaragang *dara* (as well as St'át'imcets *séna7* and the Amazonian frustratives examined by Carol & Salanova 2017). *Tɨipup*, on the other hand, blocks maximal realization of the underlying predicate, forcing the frustrated target to be identified with part or all of this predicate and ruling out proper frustrativity.

Following Carol & Salanova (2017) and Davis & Matthewson (2022), we believe that the counter-to-expectation effect of O'dam frustratives should be expressed via inertial (or stereotypical) modality. Since *tɨi*, at least, leaves open whether or not the frustrated target can still come about, non-inertia must be weakly realized: we express it as a commitment to the non-inertial status of the evaluation world (cf. Kroeger 2024). Rationally speaking, even this commitment requires a speaker to have some reason to anticipate abnormal developments, but—crucially for us—this reason need not be the definitive failure of the target of frustration.[13]

We conclude this section by summarizing our descriptive generalizations:

(17) **Two degrees of frustrativity in O'dam:**

a. **Weak** (*tɨi*): speaker commits to the non-inertial continuation of an actual event described (partially or totally) by the marked clause proposition.

b. **Strong** (*tɨipup*): speaker commits to the non-maximal realization of an event introduced in the marked clause, thereby committing to a specific type of non-inertial development for the marked-clause situation.

5 Towards a formal account of O'dam frustrativity

5.1 Antimaximality and non-inertia

We assume a *branching time framework* (Thomason 1984), in which the set of accessible *historical alternatives* $\textsc{hist}(w,t)$ of world w at time t corresponds to those

13 Looking ahead, this might reconcile the causal view of frustrativity advocated by Copley & Harley (2014: a.o.) with Davis & Matthewson's (2022) *unexpected co-occurrence*: in contexts where frustratives highlight the non-occurrence of a proposition q that is not causally linked to marked p, it may be that q provides evidence for the non-inertiality of the evaluation world.

worlds whose history is identical to w up to and including t, but which potentially diverge after t. We define the set of *inertial alternatives* as projected from a particular situation c (here, an event or collection thereof) in world w at t ($c \subseteq w$, $\tau(s) \circ t$) as that subset of $\textsc{hist}(w,t)$ in which events in s develop in the maximally causally normal (i.e., stereotypical) manner: building on Nadathur (2023), we derive a causal ordering source $\textsc{caus}(c,w,t)$ from the causal laws of a contextually-relevant *causal model* for c (see also Kaufmann 2013; Nadathur & Filip 2021).

(18) *Inertial alternatives*: $\textsc{inr}(c,w,t) := \text{Best}_{\text{CAUS}(c,w,t)}(\cap\text{HIST}(w,t))$

Section 4.3 identified two dimensions along which frustratives may vary. With respect to non-maximality, we claim that frustratives (in O'dam, and potentially in other languages) uniformly require the realization of at least some part of an eventuality description embedded in the marked clause. They may, however, vary in whether this portion can be maximal or not. Assuming that uninflected eventuality predicates are associated with an inherent (but potentially trivial) mereological structure, we define a function $\textsc{max}$ which maps an arbitrary predicate of eventualities into a predicate of maximal P-eventualities. If the mereological structure of the input predicate P is non-trivial, $\textsc{max}$ discards P-events which are proper parts of larger P-events; if P has a trivial internal structure, $\textsc{max}$ is the identity function.[14] We propose that *tɨipup* encodes the *antimaximality* requirement in (19b).

(19) a. $[\![\textsc{max}]\!] := \lambda P_{vt} \lambda e.P(e) \,\&\, \forall e'[e \sqsubset e' \rightarrow \neg P(e')]$
b. *Antimaximality for P in world w at time t:* $\neg\exists e[\textsc{max}(P)(e)(w) \,\&\, \tau(e) \circ t]$

We make the following claims about our core frustrativity types in O'dam:

(i) **Proper** frustrativity requires the maximal instantiation of an input predicate P. This reading is incompatible with *tɨipup*, given (19b).

(ii) **Incompletive** frustrativity arises when the input predicate P is *non-maximally* instantiated. This is compatible with both O'dam frustratives, and is independent of grammatical aspect.

(iii) **Avertive** frustrativity is a special case of the incompletive reading. Following Kroeger (2024), we suggest that—absent overt prospectivity—avertive readings arise when the input predicate P is coerced into an *inchoative*. As sketched in (20), $\textsc{incho}$ maps a predicate P to a telic predicate of events

14 The function $\textsc{max}$ is similar, but not identical, to Krifka's (1989; 1998) treatment of *quantization* for telic predicates. Per Krifka, a quantized predicate of events is one which only includes P-events which contain no proper subparts that also qualify as P-events, whereas we characterize maximal P-events as those that are not properly contained in a larger P-event. We select the latter formulation in order (ultimately) to allow ourselves to talk about partial (non-maximal) events as validly instantiating a predicate P, while still maintaining a notion of predicate-relative maximality.

which are *causally preparatory* for P (i.e., a dynamic predicate whose culmination condition corresponds to the initiation of a P-event).

(20) $[\![\textsc{incho}]\!] := \lambda P_{vt} \lambda e. \exists e'[e \prec e' \,\&\, \textsc{cause}(e,e') \,\&\, P(e')]$

Composition with *tɨipup* rules out the possibility of a maximal instantiation of INCHO(P), predicting the avertive reading.

Frustratives may also vary in commitment to the non-inertial status of the evaluation world. We treat strong non-inertia as *event-specific*, committing to the non-occurrence of an inertially-predicted event (cf. 11). Weak non-inertia is *non-specific*, expressing only that the evaluation world does not belong to the inertial set.

(21) **Non-inertial commitments.** Let w be a world, $c \subseteq w$ a situation, t a time:

a. w is *strongly non-inertial* iff $\exists q_{i,st}.\neg q(w) \,\&\, \forall w' \in \textsc{inr}(c,w,t)[q(w')]$

b. w is *weakly non-inertial* iff $w \notin \textsc{inr}(c,w,t)$

Proposal (22) is our first attempt at formalizing the semantics of the O'dam frustratives. We assume that frustratives take both a (potentially covert) aspectual marker—either perfective or imperfective—and an event predicate as arguments (see Section 4.3). Grammatical aspects instantiate an eventuality predicate relative to a world-time index: where predicate P is mereologically non-trivial, we assume that O'dam aspects allow both maximal and non-maximal events as valid instantiations.

(22) **Proposal for O'dam frustratives** (preliminary):

a. $[\![\textit{tɨi}(\textsc{asp}, \mathrm{P})]\!]^{c,t} := \lambda w.\textsc{asp}(P,w,t) \,\&\, \partial(w \notin \textsc{inr}(c,w,t))$

b. $[\![\textit{tɨipup}(\textsc{asp}, \mathrm{P})]\!]^{c,t} := \lambda w.\textsc{asp}(P,w,t)$
$\&\, \partial(\exists q.\neg q(w) \,\&\, \forall w' \in \textsc{inr}(c,w,t)[q(w')])$
$\&\, \partial(\neg\exists e[\textsc{max}(P)(e)(w) \,\&\, \tau(e) \circ t])$

Per (22), *tɨi* carries only a weakly inertial presupposition, making it compatible with proper as well as incompletive and avertive readings, and leaving open the possibility of a "better" outcome. *Tɨipup* imposes strong non-inertia as well as maximality, blocking proper readings and precluding the frustrative target (either a maximal realization of P or any P-event at all) in its incompletive and avertive uses.

While proposal (22) derives the desired contrasts, (22b) still falls short on our desiderata. For one, strong inertia is formulated as in (11), and our concern with the introduction of a non-explicit proposition q—i.e., that it potentially overgenerates targets for incompletive and avertive readings—is equally applicable here. Second, while (22b) does leave *tɨipup* stronger than *tɨi*, it does so both by replacing a component of (22a) as well as by adding something entirely new: this is not quite the simple parametric difference which we originally proposed, and would be difficult to maintain should it be shown that *tɨipup* is morphologically built upon *tɨi* (i.e. fn 5).

We suspect that requiring both specifications is in fact superfluous: while a formulation like (21a) is needed to express strongly non-inertial effects in the absence of antimaximality, we believe that these effects can be mediated through antimaximality. We formulated the generalization in (17b) to invoke this idea: where the target of frustration is (some part of) an event in the denotation of predicate *P*—as in incompletive and avertive readings—antimaximality alone is enough to preclude a "better" outcome. The counter-to-expectation effect of *tɨipup*, then, can be captured simply by weak inertia, resulting in the following revision: the lexical entry for *tɨipup* in (23) simply adds a clause to the semantics of *tɨi*.

(23) **Strong frustrativity in O'dam** (revised):

$$[\![\textit{tɨipup}(\textsc{asp}, \mathrm{P})]\!]^{c,t} := \lambda w.\textsc{asp}(P,w,t)\,\&\,\partial(w \notin \textsc{inr}(c,w,t)) \\ \&\,\partial(\neg\exists e[\textsc{max}(P)(e)(w)\,\&\,\tau(e)\circ t])$$

5.2 Aspect, telicity, and event maximality

We assume that the mereological structure of an event predicate *P* is determined by idiosyncratic properties of its (de)composition, such as a thematic object that can be partially affected: for instance, we take it that the predicate in (9a) can be non-maximally instantiated because one can see a given object when only part of it is visible. This accounts for the availability of *tɨipup*-marked incompletivity with some atelic predicates, as well as the lack of such readings for achievements. It does not, however, explain why accomplishments can be used incompletively (as in 6b).

Accomplishments are usually treated as mereologically trivial: they are taken to denote only culminated eventualities. This view is reconciled with the observation that accomplishment predicates can be felicitously used to describe non-culminating events—e.g., in progressive contexts (Dowty 1979)—by assuming that a wider-scope intensional operator relates actual events to qualified (culminating) *P*-events instantiated across a set of inertial alternatives. The consequence of this view for our current study is that antimaximality alone cannot license either incompletive or avertive frustrativity for accomplishment predicates: in each case, partial realization of the marked-clause event (whether *P* or INCHO(*P*)) will require an intensional aspect such as the imperfective to license treating actual but non-culminated events as initial phases (or *stages*; Landman 1992) of the target event type.

The prediction, then, is that incompletivity should only be available for accomplishments if the frustrative-marked clause is interpreted with an intensional aspect. If imperfectives (progressives) are the only such aspects, the prediction is falsified by data like (7b), which shows overt perfective marking in an avertive context. Of course, this data might be easily be explained by assigning an intensional meaning to the O'dam perfective (cf. Altshuler 2014): although we thus far lack independent motivation for such a move, we also do not see a clear reason to rule it out.

We would like, however, to suggest an alternative, building on Nadathur & Filip's (2021) recent work on the nature of telicity. Nadathur & Filip suggest that the intensionality inherent in non-culminating uses of telic predicates is not introduced by grammatical aspect, but inheres in the denotation of accomplishments themselves: the idea is that these predicates have a non-trivial mereological structure induced by a contextually-relevant causal model in which the culmination condition C_P of predicate P occurs as a dependent variable. The structure of this model describes a set of causal laws (Kaufmann 2013) which act as the ordering source for *teleologically-optimal alternatives* for C_P. On this approach, a maximal P-eventuality corresponds to a chunk of one such optimal alternative, running from the reference situation (from which alternatives are projected) up to and including to the point of culmination; partial P-events share their initiation point with maximal ones but simply terminate prior to C_P. The denotation of a predicate P in context c, then, can be conceptualized as a set of "nested temporal slices" of teleologically optimal culmination alternatives ($\mathrm{Best}_{\textsc{caus}(c,w,t)}((\cap \textsc{hist}(w,t)) \cap C_P)$): event e_1 is part of e_2 ($e_1 \sqsubseteq e_2, e_1, e_2 \in [\![\mathrm{P}]\!]^{c,w,t}$) just in case e_2 is a continuous causal development of e_1 and both are parts of some maximal (culminating) $e_3 \in [\![\mathrm{P}]\!]^{c,w,t}$ (Nadathur & Filip 2021: 620).

This proposal has two immediate consequences. First, it allows antimaximality to meaningfully interact with accomplishments, which are no longer mereologically trivial: this unifies incompletive frustrativity for telic and atelic predicates.[15] Second, grammatical aspects can now be uniformly extensional, simply constraining the relationship between an eventuality and a world-time index. If (non-)culminated events can validly instantiate a telic predicate P (as per Nadathur & Filip), both (24a) and (24b) will be compatible with non-culminated uses of such predicates:[16]

(24) a. $[\![\textsc{pfv}]\!] := \lambda w \lambda t \lambda P_{vt}.\exists e[\tau(e) \subseteq t \,\&\, P(e)(w) \,\&\, \forall e'[e \sqsubseteq e' \rightarrow \neg P(e')(w)]]$
b. $[\![\textsc{impf}]\!] := \lambda w \lambda t \lambda P.\exists e[\tau(e) \supset t \,\&\, P(e)(w)]$

Removing modality from the aspectual layer allows us to more clearly differentiate the effects of aspect from the modal effects of frustrativity. We view this as a positive outcome, since the complexity added by adopting Nadathur & Filip's approach to telicity is compensated both by a simplification in the treatment of aspect and in the conceptual aspect-frustrative distinction. However, we note that the lexical entries in (22a) and (23) derive the observed frustrative effects in O'dam on a more standard theory, in combination with selectively intensionalized grammatical aspects.

15 We can also revise our definition of INCHO: (i) uses causally-structured inertial alternatives to make INCHO(P) telic—with culmination condition P—in the sense of Nadathur & Filip:

(i) $[\![\textsc{incho}]\!]^{c,t} := \lambda w \lambda P \lambda e.\forall w' \in \textsc{inr}(c,w,t)[\tau(e) \circ t \,\&\, e \text{ in } w' \rightarrow \exists e'[\tau(e) \prec \tau(e') \,\&\, P(e')(w')]]$

16 Perfective (24a) differs from imperfective (24b) in temporal containment as well as in a *termination* requirement, which ensures that the instantiated P-event is the largest of its type at reference time.

5.3 Other uses for O'dam frustratives

Our proposals make certain predictions regarding the distribution of O'dam frustratives in optative, counterfactual and/or polite request uses. If we are on the right track, we expect that the weakness of *tɨi* will render it compatible with 'true' optativity–i.e., desires or intentions that may yet come to fruition. Strong non-inertia (weak non-inertia plus antimaximality) predicts that *tɨipup* will be incompatible with such uses, but may be permissible—even preferred—with strictly out-of-reach desires (e.g., past wants that narrowly failed to materialize). Preliminary data supports these predictions: *tɨi*-marked (25) seems optative, but *tɨipup* is so far rejected.[17]

(25) **tɨi** jix=muki-m
FRST COP=die.SG-DES
'I want to die'

We anticipate similar effects in conditionals: uncertain or less-vivid conditionals should be expressible only with *tɨi*, but strict counterfactuals should prefer *tɨipup*. Example (26) is compatible with these predictions: absent reliable information about the status of the antecedent, the "frustrated" status of the consequent is indeterminate, and we find that *tɨi* but not *tɨipup* is accepted here.

(26) No'=pimɨ-t tu-jimchuda añ xib {**tɨi/#tɨipup**}
COND-2PL.SBJ-PFV DUR-plough 1SG.SBJ now {FRST/#FRST.NONMAX}
tu-ɨ'xcha-'
DUR-plant-IRR
'If you all ploughed earlier (I don't know if you did), I could plant now.'

Finally, *tɨi* should be felicitous as a marker of polite requests, since it lets the addressee select between acquiescence and non-cooperation. Since *tɨipup* prefigures non-cooperation, we predict that it cannot be used this way. Examples (27a)-(27b) seem to support this view: (27a) is perceived as a request, but (27b) instead explains why the speaker is *not* making a request.

(27) a. **tɨi**=ñ jix=a' na=p gɨt
FRST=1SG.SBJ COP=want SUB=2SG.SBJ CONTR
ba-ñ-bhɨidhika-' gu tɨm
CMP-1SG.PO-bring-IRR DET tortilla
'Please bring me tortillas.' (*lit:* I want you to bring me tortillas *tɨi*)

b. **tɨipup**=añi-ch jum-taa gu tɨm dai
FRST.NONMAX=1SG.SBJ-PFV 2SG.PO-request.PFV DET tortilla but
na=Ø-t cham jir=aich
SUB=3SG.SBJ-PFV NEG COP=arrive.APPL.PFV

17 Although it may not occur on the expected or intended timeline, dying can never be entirely precluded.

'I would ask that you bring tortillas but you didn't before.'

This discussion is preliminary: we continue to investigate the full range of O'dam frustrativity, and leave a definitive assessment of data like (25)-(27) for future work.

6 Conclusions and typological perspective

Regardless of the language of study, the semantic analysis of frustrativity poses a dual challenge: we must explain both observed variation in how much of a marked-clause event is actually realized, as well as finding a principled way of linking this event to a target of frustration which may be neither lexically determined nor overtly expressed. The distribution of frustrative readings in O'dam adds an additional layer of complication: as discussed in Section 4, existing analyses either allow us to explain incompletive and avertive readings (to the exclusion of proper frustrativity) or link the distribution of frustrativity type to grammatical aspect in a way that does not map cleanly onto the O'dam data. We also found that the treatment of non-inertial commitments in existing analyses was insufficiently granular to capture the observed difference between *tɨi* and *tɨipup*.

Our solution was to propose that O'dam frustratives vary both with respect to the strength of non-inertial commitments as well as with respect to an event-based notion of maximality. Our analysis suggests that frustratives cannot always be treated as purely propositional operators, but may merge with the aspectual layer to introduce modal considerations of the sort typically linked to *non-culmination*.

Looking ahead, we predict the existence of at least three types of frustratives, each of which has been attested: weakly inertial but MAX-compatible particles with the usage profile of O'dam *tɨi* and Kimaragang *dara* (Kroeger 2024), strongly inertial but MAX-compatible frustratives such as Tohono O'odham *cem* (Copley & Harley 2014) and St'át'imcets *séna7* (Davis & Matthewson 2016, 2022), and a third, new category of weakly inertial but *antimaximal* markers like O'dam *tɨipup*. Our suggestion in Section 5.1 was that an antimaximality requirement obviates the conceptual difference between strong and weak inertia, so that we do not predict the existence of a distinct fourth category of strongly inertial and antimaximal particles. This remains to be properly investigated. We see this as a particularly rich area for future investigation, since—if no such fourth class can be found—one might also reasonably ask whether strong non-inertia is indeed truly attested, or perhaps arises via default pragmatic strengthening of weak non-inertia in languages where a particle with the profile of *tɨi* is not paired with a stronger alternative.

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